DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Staff answers · model and retrieval

Source authority and prior learning

Recommended generic S4: Describe DNA, genes & chromosomes. Exact goal and criteria above are unchanged. Source: Launch/Autumn_2_2026-27/SCI_LAUNCH_A2_W04/SCI_LAUNCH_A2_W04.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d. Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B42 Aut2·W4, C42 Describe DNA, genes & chromosomes.

Starter keys in order: (1) To check whether it benefits people and what risks it has [Autumn 2 Week 3]. (2) Each new daughter-cell nucleus [Autumn 2 Week 1]. (3) In the nucleus [Autumn 1 Week 1]. “Last term” refers to the workbook Autumn 1 block. Confirm actual taught coverage rather than inventing a prior lesson.

Full model, check and fade

Mark a short section of a DNA cord as a gene. Coil the cord as one chromosome; place it inside a border for the genome. Name the section, molecule, package and complete DNA set.

Show-me answer: A gene is a section of DNA. Hear both responses through pointing, board or card and re-model the named step if insecure. The clip is a marker; the DNA section beneath it is the gene and codes for a specific protein. A chromosome is a long DNA molecule packaged with proteins; the genome is all an organism’s DNA, including other DNA.

The nested diagram and string are simplified, not to scale. The genome border is the full DNA set, not a whole-cell label. No personal medical claim is made. Full model → missing gene label at We do → no filled answer at Independent. Next lesson hide the genome label and reconstruct the whole-set meaning before genetic-cross work.

Misconceptions

Chromosome = gene: uncoil the cord, point to one marked DNA section, then to the whole DNA package. Gene = protein or separate cell: the clip marks DNA beneath; that section codes for a protein. Do not say that the clip itself is a gene.

A genomics laboratory scientist compares gene sections and chromosome changes when interpreting a test. This career example is not a medical claim about a learner.

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Staff answers · shared work and hinge

Missing step: gene; the short marked DNA section codes for a specific protein. Leave the pupil blank unfilled. Speaking frame: “A gene is part of …; a chromosome contains …”

Match keys in source card order: short coloured DNA section → gene; whole genetic molecule → DNA; long DNA package → chromosome; all of the organism’s DNA → genome. Option order remains DNA / chromosome / gene / genome. Explain one mismatch and allow retry; do not store answers in HTML.

We do Supported/F: place gene on the short marked part of the full cord. Independent/M: explain how this short section relates to one gene and the larger package. Aspire/H: one chromosome = one gene confuses scale and role. These are alternative support routes, not extra consecutive tasks.

Four-option hinge

Correct B, the second option: A gene within the DNA molecule. A treats a short DNA region as the whole genome. C labels a DNA region as a whole cell and overlooks DNA. D calls a gene a chromosome and confuses one chromosome with all DNA. Keep source question and option order. There is no separate three-choice step check in this source.

Pair adaptation of the source class threshold: hear BOTH learners’ own choices and reasons. Continue to transfer when both are secure. If either is unsure, re-model their named step and ask the parallel hinge while the other continues. Do not use a literal three-in-four pupil count or infer a grade.

Parallel hinge: “A second diagram outlines an organism’s complete DNA set. Which label fits: gene or genome?” Answer: genome; it includes all its DNA.

Independent keys

Comparison Model A is accurate: gene marks the short DNA section. Model B reverses gene and DNA; correct the labels and justify with section/whole molecule. Supported labels gene, DNA, chromosome, genome. Independent explains gene is DNA, not another material. Aspire corrects the gene label and names its protein instruction. Go deeper keeps gene section, DNA molecule, chromosome package and entire DNA set distinct. Two-diagram audit: (1) gene is part of DNA, not DNA inside a gene; (2) one chromosome is part of the genome, not the entire genome. Justify both.

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Staff answers · invitations, prompts and evidence

Optional invitation bank · pass is valid

Route	Invitation	Expected answer
Supported	What is a gene part of?	DNA.
Supported	What is the name for all an organism's DNA?	The genome.
Independent	How does a chromosome relate to genes?	A long packaged DNA molecule with gene sections.
Independent	What does the gene section code for in this specification?	A specific protein.
Aspire	Why is the whole chromosome not correctly labelled as one gene?	A gene is only a DNA section, with a coding function.
Aspire	What is lost when genome means one chromosome?	Genome is the entire DNA, not one package.

Use invitations privately without cold calling. Each learner may plan in any language and point, sign, speak, draw, type, write, use AAC, an exact scribe or staff-captured photo. No required personal or family disclosure.

Held five-step prompt ladder · staff only

1. Point to the relevant place / instruction.
2. Ask one short question: "What is it asking you to find?"
3. Rephrase the instruction in simpler language.
4. Provide a scaffold or limited choice.
5. Model the first step only, then return the task to the pupil.

First wait five seconds for an independent label. Source cue examples within the ladder: point to the marked string; ask “Is this marker the entire string or part of it?”; model the DNA label on one string while the learner labels the coloured part; complete one label only if needed, then offer a fresh diagram alone. No physical prompting. Record the deepest step separately from the pupil I · S · C box.

Stage scripts: I do “Point to the short marked section, then trace the whole DNA string.” We do “Where is the missing label on this model? What size is it?” Independent “Show the incorrect label first. What does the short section stand for?”

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Staff answers · access, marking and targets

Access and regulation

Cognition: enlarged full cord beside an unlabelled model; fade only the gene marker. Communication: rehearse “part of” with a partner; allow point/sign/AAC labelled explanation. Regulation: private heads-up before faulty-model critique, pause choice and one-label return. Sensory/physical: high-contrast cord, easy-grip clip or digital labelled image/speech-to-text. Language: the same four cards with meanings/examples and any-language planning; keep Independent/Aspire science demand. EAL is language, not SEN.

Check in before hinge, the top rung and Independent. Allow space and one return step; resume at the last completed step with one small choice and private correction. No evidence during distress. Use actual staffing; one learner explains while the other asks/locates, then swap. Keep separate individual work.

Mark actual evidence

Use VF · WS · I · NS(+) · E · R · // · ? only when they occur. R requires the next-step response heard and judged next lesson; it is never automatic. Hear actual Review feedback, invite a real self-edit // or requested read-back ?, and hear the Exit model/reason and precise next sentence. Record hinge split, misconception, actual code and deepest prompt separately.

Book-first: use the exercise book or agreed response mode; do not require writing on these resource pages. Evidence is an observation/result table OR labelled explanation, Review edit and Exit alongside the original. Do not invent a result table for this modelling lesson. Label spoken/practical/photo evidence SCI-LAUNCH-A2-W04 on [SYS]; no pupil data in HTML and no pupil phones needed.

Individual targets

Missing gene: sequencing, attention and independence plus actual prompt. Hinge: agreed response mode/turn-taking without mandatory speech. Arrival/pause: private engagement. Independent: accessible input and target-related independence, not diagnosis. Record actual four-word first-language rehearsal. Link E to each learner’s own education/support/health/care/personal targets on [SYS]. Follow the individual behaviour/support plan where it differs.

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Staff answers · safety, mapping and handoff

Optional prepared-model rehearsal

Kit: short DNA cord, large blunt clips, genome/chromosome frames, tray. Mark one gene with a blunt clip and nest the cord in its frames. Risk: clip pinch and loose cord catch. Use large blunt clips, short cord on a tray, away from necks. Mandatory controls: short length only, counted out and back in, adult-supervised, removed at once if a learner is dysregulated. Confirm actual kit and school risk assessment. Offer a digital labelled picture and no forced handling. This is not a core practical.

Mapping and scheme context

[AWARD] [QUAL] Foundation Biology/Combined Science Paper 1: shared points 3.5 (genome/gene) and 3.13 (chromosome/gene). No Biology-only or higher-tier point is assumed; workbook approach is not an extra specification point. Scheme evidence: [AWARD] [QUAL] foundation topic tests and science units; formative topic quizzes/practical write-ups; summative Topics 1–3 assessment in December.

Evaluation and next lesson

Date taught _____ Delivered as planned _____ Criteria met _____ Adaptations and reasons _____ Change/coverage impact _____ Evidence banked _____. Printed Date and Version stay blank. Record response mode, independence and deepest prompt in handoff.

If behind, retain accurate-model choice plus reason, individual evidence and safety. After a secure hinge shorten repeated oral rehearsal. If necessary merge Exit reflection into Autumn 2 Week 5 Arrival, recording reason and coverage impact. Do not replace individual work with group output. Pass checked understanding and unresolved errors to the next starter.

Next weekly line: Complete a genetic cross (digital Punnett square). Hide the genome label for the next model and reconstruct the whole-set meaning first.

Source resets: one minute inside We do 7, Independent 8 and Review 4. Stage budget: Arrival 5, Starter 3, I do 5, We do 7, We do 2 5, Independent 8, Review 4, Exit 3 = 40 minutes. Optional resources and alternative route views add no minutes.

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